# ECON PhD Math Bootcamp 2025

**Syllabus**

## Course Overview

This math bootcamp is designed to prepare incoming PhD students in economics with the essential mathematical tools and intuition required for first-year coursework in microeconomics, macroeconomics, and econometrics. The focus is on refreshing and strengthening foundational concepts in algebra, calculus, linear algebra, optimization, probability, and dynamic modeling, with an emphasis on applications relevant to economics.

## Course Structure

- **Format:** Lecture + problem-solving practice
- **Dates:** July 28 (1-4pm), July 29-Aug 14 (9am-1pm daily), Aug 15 (Exam)
- **Location:** GC 5490
- **Breaks:** 10–15 minutes halfway through each session
- **Assessment:** Diagnostic quiz (July 28), problem sets, and final exam (Aug 15)
- **Contact:** ayixunbati.nuerlantai@utah.edu

## Learning Objectives

By the end of this course, students will be able to:

- Solve optimization problems using calculus and linear algebra techniques.
- Understand and manipulate functions in single and multivariable settings.
- Work confidently with matrices, vector spaces, and eigenvalue problems.
- Apply basic dynamic systems concepts (difference equations, phase diagrams).
- Understand probability distributions and work with joint distributions.
- Be prepared for the math demands of core PhD courses, especially Micro, Macro, and Econometrics.

## Suggested Textbooks

You are not required to purchase a textbook, but the following are recommended by the professors who will be teaching you in your first year as references (you can find the pdf version from the shared folder on UBox):

- Simon & Blume (1994) – *Mathematics for Economists* (by Subha and Ellis)

- McKenna, Livernois & Hoy (2001) – *Mathematics for Economics, 2nd Edition* (by Ellis and Codrina)
- Wainwright & Chiang (2004) – *Fundamental Methods of Mathematical Economics, 4th Edition* (by Fancisco)
- Sydsaeter & Hammond (1995) – *Mathematics for Economic Analysis* (by Lozada)
- Prof. Lozada's Prerequisite Handout: `https://content.csbs.utah.edu/~lozada/Microeconomic_Theory_I/Handouts/prereq.pdf`

## Day-by-Day Schedule

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### Day 1: July 28 (1–4pm) – Assessment & Foundations

- **Opening & Assessment (1:00–1:30pm)**
  - Anonymous skills assessment based on Prof. Lozada's handout pages 1, 3, 4
  - Identify knowledge gaps in the cohort
- **Basic Mathematical Foundations (1:30–3:00pm)**
  - Set theory and notation
  - Function notation: $f : \mathbb{R}^n \to \mathbb{R}^1$
  - Domain and range concepts
  - Summation notation and double sums
- **Review & Questions (3:00–4:00pm)**
  - Address assessment results
  - Preview of upcoming topics

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### Day 2: July 29 – Linear Algebra Fundamentals

- **Matrix Operations (9:00–10:30am)**
  - Matrix definition and basic operations
  - Matrix multiplication and transposition
  - Symmetric matrices
- **Break (10:30–10:45am)**
- **Advanced Matrix Operations (10:45am–12:00pm)**
  - Determinants and inverse matrices
  - Matrix rank and linear independence
- **Practice Problems (12:00–1:00pm)**

### Day 3: July 30 – Linear Algebra Applications

- **Systems of Linear Equations (9:00–10:30am)**
  - Gaussian elimination
  - Cramer's rule
- **Break (10:30–10:45am)**
- **Eigenvalues and Eigenvectors (10:45am–12:00pm)**
  - Characteristic equations
  - Diagonalization
- **Economics Applications (12:00–1:00pm)**
  - Input-output models
  - Markov chains

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### Day 4: July 31 – Calculus Review I

- **Single Variable Calculus (9:00–10:30am)**
  - Limits and continuity
  - Derivatives and rules
- **Break (10:30–10:45am)**
- **Applications of Derivatives (10:45am–12:00pm)**
  - Optimization problems
  - Elasticity in economics
- **Integration Basics (12:00–1:00pm)**
  - Fundamental theorem of calculus
  - Basic integration techniques

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### Day 5: August 1 – Multivariable Calculus

- **Partial Derivatives (9:00–10:30am)**
  - Concept and notation
  - Higher-order partial derivatives
- **Break (10:30–10:45am)**
- **Gradient and Directional Derivatives (10:45am–12:00pm)**

  - Gradient vector
  - Chain rule for multivariable functions
- **Applications in Economics (12:00–1:00pm)**
  - Marginal products and rates of substitution

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### Day 6: August 4 – Optimization Theory

- **Unconstrained Optimization (9:00–10:30am)**
  - First and second-order conditions
  - Hessian matrix
- **Break (10:30–10:45am)**
- **Constrained Optimization (10:45am–12:00pm)**
  - Lagrange multipliers
  - Kuhn-Tucker conditions
- **Economic Applications (12:00–1:00pm)**
  - Utility maximization
  - Cost minimization

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### Day 7: August 5 – Concavity and Convexity

- **Convex Sets (9:00–10:30am)**
  - Definition and properties
  - Convex hull
- **Break (10:30–10:45am)**
- **Concave and Convex Functions (10:45am–12:00pm)**
  - Definitions and tests
  - Second-order conditions
- **Applications (12:00–1:00pm)**
  - Producer and consumer theory
  - Checking concavity/convexity conditions

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### Day 8: August 6 – Probability and Statistics

- **Probability Basics (9:00–10:30am)**

  - Probability density functions
  - Cumulative distribution functions
- **Break (10:30–10:45am)**
- **Joint Distributions (10:45am–12:00pm)**
  - Joint probability distributions
  - Marginal and conditional distributions
- **Applications (12:00–1:00pm)**
  - Expected value and variance
  - Proof by contradiction techniques

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### Day 9: August 7 – Differential Equations I

- **First-Order Differential Equations (9:00–10:30am)**
  - Separable equations
  - Linear first-order equations
- **Break (10:30–10:45am)**
- **Applications to Economics (10:45am–12:00pm)**
  - Growth models
  - Market dynamics
- **Solution Methods (12:00–1:00pm)**
  - Integrating factors
  - Substitution methods

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### Day 10: August 8 – Differential Equations II

- **Second-Order Differential Equations (9:00–10:30am)**
  - Homogeneous equations
  - Characteristic equations
- **Break (10:30–10:45am)**
- **Systems of Differential Equations (10:45am–12:00pm)**
  - Matrix methods
  - Eigenvalue approach
- **Phase Diagrams (12:00–1:00pm)**
  - Stability analysis

  - Jacobian matrices

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### Day 11: August 11 – Dynamic Systems

- **Discrete Dynamic Models (9:00–10:30am)**
  - Difference equations
  - Stability conditions
- **Break (10:30–10:45am)**
- **Continuous Dynamic Models (10:45am–12:00pm)**
  - Continuous-time optimization
  - Steady states
- **Applications (12:00–1:00pm)**
  - Economic growth models
  - Business cycles

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### Day 12: August 12 – Optimal Control Theory

- **Hamiltonian (9:00–10:30am)**
  - Setting up the Hamiltonian
  - Necessary conditions
- **Break (10:30–10:45am)**
- **Optimal Control Problems (10:45am–12:00pm)**
  - Current value Hamiltonian
  - Transversality conditions
- **Economic Applications (12:00–1:00pm)**
  - Optimal consumption/saving
  - Resource economics

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### Day 13: August 13 – Advanced Topics

- **Inverse Functions (9:00–10:30am)**
  - Existence and uniqueness
  - Implicit function theorem
- **Break (10:30–10:45am)**

- **Advanced Probability (10:45am–12:00pm)**
  - Moment generating functions
  - Limit theorems
- **Integration Review (12:00–1:00pm)**
  - Advanced integration techniques
  - Applications to probability

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### Day 14: August 14 – Review and Practice

- **Comprehensive Review (9:00–10:30am)**
  - Key concepts from microeconomics prep
  - Macroeconomics mathematical tools
- **Break (10:30–10:45am)**
- **Practice Problems (10:45am–12:00pm)**
  - Mixed problems from all topics
  - Exam-style questions
- **Q&A Session (12:00–1:00pm)**
  - Address remaining questions
  - Exam preparation tips

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### Day 15: August 15 – Final Exam

- **Final Assessment**
  - Comprehensive exam covering all topics
  - Focus on application to economics

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## Special Considerations

- Micro Focus: Emphasize optimization, concavity/convexity (Days 6-7)
- Macro Emphasis: Strong coverage of differential equations and dynamic systems (Days 9-12)
- Econometrics Prep: Include probability and statistics foundations (Day 8)

## Policies and Expectations

- Attendance is strongly encouraged due to the fast pace.
- Participation is encouraged—ask questions, help peers, and engage actively.
- Homework will be optional but reviewed in class.
- Final Exam is diagnostic and formative—it will help you identify your strengths and weaknesses before the semester begins.

## Optional Supplementary Materials

- Prof. Lozada's old exam problems — useful for additional practice on prerequisite math topics.
- *Linear Algebra for Dummies* — a simple refresher on linear algebra basics.
- Selected slides from Prof. Rada on dynamic models, the Hamiltonian, and Optimal Control Theory (OCT). These are optional, especially since Macro II will be taught by Prof. Erturk this year.
- Visualization tools like Mathematica, Python, or R for exploring matrices and functions.

## Recommended Learning Tools

- **Paper Reading:** Zotero — for organizing and managing references.
- **Note-Taking:** Obsidian (especially useful when linked with Zotero).
- **Writing:** Learn LaTeX for academic writing. A good starting guide is Oetiker's *The Not So Short Introduction to LaTeX2e*.
- **University Resources:** Cloud storage (OneDrive, Box, Adobe Creative Cloud), Microsoft Office 365, Grammarly (premium access with university account), access to over 200 software programs, and printing/scanning services.
- **Equipment Checkout:** Laptops (Mac or Windows) and other equipment can be borrowed through the library by submitting a request form. More details at this page.
- **AI Tools:** Claude, ChatGPT — for brainstorming, summarizing, or language help.